



TXW80X audio solution development guide



Notice

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Zhuhai Taixin
Semiconductor Co.,Ltd

3rd Floor, Building 11, Science and Technology Innovation Park, No. 1, Gangwan, High-tech Zone, Zhuhai City

Confidentiality level	A	TXW80X SDK Quick Start Manual	File No	TX-0000
Release date	2022-08-08		file version	V1.1

Revision history

date	Version	describe	Revised by
2022-08-08	V1.1	Add I2S development guide	TX
2022-05-15	V1.0	initial version	TX



Zhuhai Taixin Semiconductor Co., Ltd.
Zhuhai Taixin Semiconductor Co.,Ltd

3rd Floor, Building 11, Science and Technology Innovation Park, No. 1, Gangwan, High-tech Zone, Zhuhai City

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		Zhuhai Taixin Semiconductor Co.,Ltd	3rd Floor, Building 11, Science and Technology Innovation Park, No. 1, Gangwan, High-tech Zone, Zhuhai City	

1 Overview

This article mainly describes the video development process.

This document is mainly applicable to the following engineers:

- Technical support engineer
- Solution software development engineer

Product scope applicable to this document:

model	encapsulation	Package
TXW806	QFN56/48/40	

2. Hardware development board

For quick start and solution evaluation, we provide development boards for various application scenarios.

2.1. Audio and video development board

2.1.1. Audio and video development board interface introduction

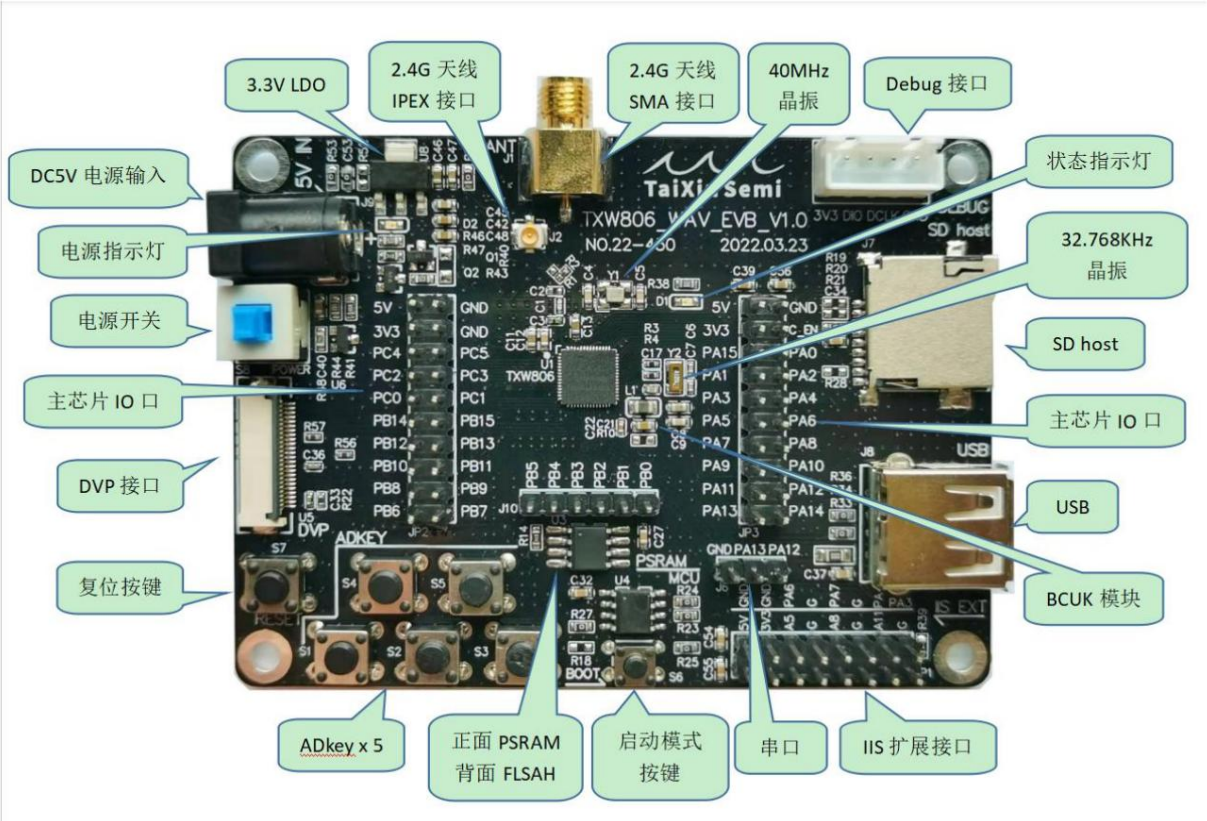


Figure 3.1.1.1 - Main view of audio and video development board

Special Instructions

Mode start button: This button can save the system with one click, run to death when the chip is powered on, cklinc programming and other upgrades

Used when all levels are disabled.

IIS extension interface: This interface is used to expand the audio daughter board.

3. Audio development configuration process

3.1. Audio solution related configurations

The solution configuration mainly depends on the project_config.h configuration. The following configurations are all based on the SDK-based image transmission protocol.

```
#define USB_HOST_EN          1
#define MACBUS_USB

#define USB_EN              0
#define DVP_EN              0
#define JPG_EN              (1 && DVP_EN)
#define SDH_EN              1
#define FS_EN               1
#define VCM_EN              (0 || DVP_EN)

#define FRAME_MANAGEMENT    1
#define OPENDML_EN          1
#define UART_INPUT_EN       1
#define UART_FLY_CTRL_EN    0
#define PWM_EN              0

#define AUDIO_EN             (0 && FRAME_MANAGEMENT)
#define IIS_AUDIO_EN         (0 && FRAME_MANAGEMENT)
#define RTP_SOUND            (1 && (AUDIO_EN || IIS_AUDIO_EN))

|
#define MJPEG_VIDEO          (1 && FRAME_MANAGEMENT && OPENDML_EN && FS_EN && SDH_EN && JPG_EN) //基于框架的mjpeg录像
#define UVC_VIDEO            (1 && FRAME_MANAGEMENT && OPENDML_EN && FS_EN && SDH_EN && USB_EN) //基于框架的uvc录像
```

1. USB_EN starts the USB camera (usb and dvp cannot coexist temporarily)
2. DVP_EN starts the dvp camera (usb and dvp cannot coexist temporarily)
3. SDH_EN and FS_EN are sd card and file system initialization
4. FRAME_MANAGEMENT (only supports SDK's image transmission protocol) framework startup, when plugging in psram, you need to use
able
5. OPENDML_EN needs to be activated when recording is required (only supports the recording format of SDK, other formats need to be
accomplish)

3.2. PDM Microphone

3.2.1. PDM audio related parameter configuration

There is a demo with built-in audio in the sdk, which is mainly opened through the macros #define AUDIO_EN and #define FRAME_MANAGEMENT (project_config.h). The default is to adapt to the sdk, which can transmit image and audio together. If you need to customize the protocol, Need to develop by yourself

3.2.2. PDM audio interface description

The audio interface of pdm has been encapsulated in the sdk. You need to register the corresponding interface. You only need to pay attention to the following interfaces, as shown in the figure:

```

7:
8: void    pdm_audio_close(void *audio_hdl);
9: int     pdm_audio_start(void *audio_hdl);
0: void    pdm_audio_register(void *audio_hdl, void *priv_el, int play_size, set_buf set_buf, get_buf get_buf);
1: void    *pdm_audio_open(enum pdm_sample_freq freq, enum pdm_channel channel);
2:
3:

```

1. pdm_audio_open opens the pdm hardware interface . Parameters are audio sample rate and channel . Reference enumeration

pdm_sample_freq and pdm_channel, the return is a pdm audio handle (need to be used later)

2. pdm_audio_register registration interface, parameters: audio_hdl is the handle of pdm audio, priv_el is private

There is a structure (register by yourself, optional, see application layer implementation), play_size sets the length of audio playback (configuration

cannot be modified after setting, otherwise an error occurs), set_buf (must be registered) returns the buf address of the recording, get_buf (must be registered)

Registration required) After the recording is completed, the corresponding buf is returned or the structure pointer of buf can be found through

Parameters are passed in (refer to the implementation of audio_demo.c)

3. pdm_audio_start recording officially starts, the internal implementation will call set_buf

4. pdm_audio_close closes pdm recording

Note: The main interfaces implemented are the two functions set_buf and get_buf. priv_el is used as an auxiliary function. set_buf and

get_buf is basically an interrupt call, so the implementation cannot be a time-consuming operation. Time-consuming operations can be placed in threads.

3.2.2.1. PDM recording demo part implementation code reference (audio_demo.c)

1. Function implementation of set_buf, as shown in the figure:

```

//配置录音的buf地址
static void *audio_set_buf(void *priv_el, void *el_point)
{
    struct audio_frame_priv *priv = (struct audio_frame_priv *)priv_el;
    struct list_head* get_analy_node;
    struct list_head **point = (struct list_head**)el_point;
    void *buf = NULL;
    get_analy_node = get_app_analyze_node(&(priv->pdm_app));
    if(get_analy_node)
    {
        buf = get_real_buf_adr(get_analy_node);
        if(!buf)
        {
            *point = NULL;
            return NULL;
        }
        *point = get_analy_node;
        set_app_buf_len(get_analy_node, AUDIOLEN);
    }
    return buf;
} « end audio_set_buf »

```

priv_el is a private structure, and all buf applications are obtained from the private structure.

audio_set_buf returns a buf address, which is used to fill in the recording data.

*el_point is a pointer reference and must be assigned a value. This can be buf. The address can be other addresses. Finally, it will be returned

Back to get_buf, what is returned here is get_analy_node, get_buf will get get_analy_node, and index to the buf address through get_analy_node, so get_analy_node is bound to the buf address

2. Function implementation of get_buf, as shown in the figure:

```

//获取录音的buf地址
static void audio_get_buf(void *priv_el, void *el_point)
{
    struct audio_frame_priv *priv = (struct audio_frame_priv *)priv_el;
    struct list_head *get_analy_node = (struct list_head *)el_point;
    int res;
    if(!get_analy_node)
    {
        printf("%s:%d err\n", __FUNCTION__, __LINE__);
        return;
    }
    res = csi_kernel_msgq_put(priv->pdm_msgq, &get_analy_node, 0, 0);
    //正常应该保证不进这里,如果进来代表任务没有获取队列,直接配置下一个buf导致的
    if(res)
    {
        printf("%s:%d err res:%d!!!!!!!!!!!!!!!!!!!!\n", __FUNCTION__, __LINE__, res);
        force_del_frame(get_analy_node);
    }

    return;
}
} « end audio_get_buf »

```

priv_el is a private structure

*el_point is the address of get_analy_node. Refer to set_buf, which is the address where the recording buf can be found. Here, get_analy_node is sent through the message queue and called in other threads, because set_buf and get_buf are both interrupt callbacks. Therefore, it is not suitable to do some audio processing or time-consuming operations. res represents whether the message queue is successfully sent. If it is not successful, get_analy_node will be deleted and the resources will be released.

3. Audio processing task implementation, as shown in the figure:

```

static void audio_deal_task(void *arg)
{
    struct audio_frame_priv *priv = (struct audio_frame_priv *)arg;
    int16* addr;
    int res;
    int i;
    struct list_head* get_analy_node;
    TYPE_FIRST_ORDER_FILTER_TYPE filter_ctl;
    rm_dc_filter_init(&filter_ctl);
    while(1)
    {
        res = csi_kernel_msgq_get(priv->pdm_msgq, &get_analy_node, -1);
        if(!res)
        {
            //中断有get_analy_node传出来,代表可以发送,如果是NULL,则去设置下一个buf
            if(get_analy_node)
            {
                #if 1
                if(!priv->stable_flag)
                {
                    priv->stable_num--;
                    if(priv->stable_num <= 0)
                    {
                        //等待pdm设备稳定,所以将buf重新给到pdm硬件使用
                        priv->stable_flag = 1;
                    }
                    force_del_frame(get_analy_node);
                    continue;
                }
                #endif

                //将音频数据滤波处理
                addr = (int16*)get_real_buf_addr(get_analy_node);
                for(i=0; i<AUDIOLEN/2; i++) {
                    addr[i] = rm_dc_filter(&filter_ctl, addr[i]);
                    addr[i] = pcm_volum_gain(addr[i], PCM_GAIN);
                }

                //设置处理完毕的时间戳
                set_real_buf_timestamp(get_analy_node, csi_kernel_get_ticks());
                printf("N");
                map_realnode_2_app_node_msg(&(priv->pdm_app), get_analy_node);
                wake_up_analyze_list_app(&(priv->pdm_app));
            }
        }
    }
}
} « end if get_analy_node »
} « end if !res »
else
{
    printf("%s:%d err @!!!!!!!!!!!!\n", __FUNCTION__, __LINE__);
}
} « end while 1 »
} « end audio_deal_task »

```

Since the obtained pdm audio has not been filtered, the audio buf is passed through the message queue in get_buf.

Sent to the audio_deal_task task, so all audio data will be processed here. 1 First get get_analy_node, and then

- ① get the actual recording buf address through get_real_buf_addr (in set_buf has already explained that get_analy_node and buf are a binding function)
- ② Buf data is processed through rm_dc_filter and pcm_volum_gain, and the final result is pcm data

4. Demo initialization process:

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```

9. `buf_size` is the size of `play_empty_buf`

10. `set_buf` and `get_buf` are callback functions for playing audio or recording. They are very important. For parameter description, please refer to [the example in 3](#).

illustrate

11. `priv_el` is a private structure, the first parameter of `set_buf` and `get_buf`. If not needed, set it to `NULL`.

It is for the use of `set_buf` and `get_buf`

```

24
25 //在有mic和喇叭的时候, sample_freq与sample_bit应该要一致, 所以只有一个是有值, 默认是以喇叭的配置为准
You, 13分钟前 | 2 authors (oudi and others)
26 typedef struct audio_i2s_config
27 {
28     //无需用户配置
29     struct i2s_device *i2s_dev;
30     void *current_node;
31     void *reg_node;
32     i2s_irq_func irq_func;
33
34     //用户配置
35     uint8 type; //0:喇叭 1:麦克风
36     int i2s_dev_id;
37     int sample_freq;
38     int sample_bit;
39     int data_fmt; //I2S_DATA_FMT_I2S I2S_DATA_FMT_MSB I2S_DATA_FMT_LSB I2S_DATA_FMT_PCM
40     void *play_empty_buf; //作为喇叭的时候, 需要配置, size与buf_size一致, 可以是malloc也可以是固定, 如果是malloc, 需要自己去free
41     int buf_size;
42     audio_i2s_set_buf set_buf;
43     audio_i2s_get_buf get_buf;
44     void *priv_el;
45
46 } audio_i2s_config;
47

```

2. As shown in the figure: 8211 initialization

```

*****/
audio_8211_frame_priv * audio_8211_demo_init(const char *name)
{
    int ret;
    audio_i2s_config *play = NULL;
    void *empty_buf = NULL;
    play = (audio_i2s_config*)malloc(sizeof(audio_i2s_config));
    empty_buf = (void*)malloc(PLAY_EMPTY_SIZE);
    if(!play || !empty_buf)
    {
        printf("%s:%d\n", __FUNCTION__, __LINE__);
        goto audio_8311_demo_init_err;
    }
    memset(play, 0, sizeof(audio_i2s_config));
    memset(empty_buf, 0, PLAY_EMPTY_SIZE);

    //OS_TASK_INIT("sim_mic", &sim_mic_task, sim_mic_thread, (uint32)NULL, OS_TASK_PRIORITY_BELOW_NORMAL, 1024);
    //os_sleep_ms(200);

    //该任务及资源初始化
    audio_8211_frame_priv *audio_8211_priv = audio_8211_creat(name);
    //8311 install, 注册对应函数
    //对play mic config初始化
    play->type = PLAY_MODE;
    play->i2s_dev_id = HG_IIS0_DEVID;
    play->sample_freq = I2S_SAMPLE_FREQ_16K;
    play->sample_bit = I2S_SAMPLE_BITS_16BITS;
    play->data_fmt = I2S_DATA_FMT_LSB;
    play->priv_el = audio_8211_priv;
    play->set_buf = audio_play_set_buf;
    play->get_buf = audio_play_get_buf;
    play->buf_size = PLAY_EMPTY_SIZE;
    play->play_empty_buf = empty_buf;
    int res = audio_i2s_install(play, NULL);
    printf("%s:%d\n", __FUNCTION__, __LINE__);
}

```

Mainly after configuring some necessary parameters of play

Call `audio_i2s_install` to run normally

`audio_i2s_install` has two parameters, namely playing audio and microphone. 8211 only plays audio, so the main

Just register to play

3. Next, analyze the meaning of `set_buf` and `get_buf`

parameters. In the demo, `set_buf` corresponds to `audio_play_set_buf`, and `get_buf` corresponds to `audio_play_get_buf`.

`set_buf (void *priv_el, void *el_point, int *buf_size)`, `priv_el` corresponds to `priv_el` in `audio_i2s_config`. The return value is the `buf` that needs to be configured. `*buf_size` is the size of the `buf` returned by the value. It is returned through pointer assignment and is used to set the playback audio. The size or the recording size, `*el_point` is a pointer. After the interrupt is completed, it will be passed to `get_buf`. The return value and `*buf_size` are necessary. `priv_el` and `*el_point` are auxiliary to obtain the information of `buf` and `buf_size`.

`get_buf (void *priv_el, void *el_point, int *buf_size)`; `priv_el` corresponds to `priv_el` in `audio_i2s_config`, `*el_point` corresponds to `*el_point` in `set_buf`, what content is assigned by `set_buf`, what content can be obtained here, `*buf_size` is not called yet (Keep consistent with `get_buf` parameters)